DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2022/2023

THIRD YEAR **SEMESTER** 2 SUPPLEMENTARY/ SPECIAL EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
CIT 3203: RESEARCH METHODOLOGY

DATE: **TIME:**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

Retaining employees in the Telecommunication industry has been a problem. Most employees who work for these companies are young talented people and over years there can be an observed trend of employees leaving and seeking jobs from other companies, some of other industries. This problem does not require an immediate solution and hence requires adequate time for investigation. Tom who is an employee of one of the telecommunication company, out of self will starts a research work to find methods of retaining employees for the companies in this industry. The research work must be extended for a period of five years so that his research can validate if the five-year period has an impact on his results. He intends to collect his data from all telecommunication companies from three countries Kenya, Uganda and Tanzania but unfortunately, he discovers that it will not be an easy process because most employees have the fear of loosing their jobs by providing information about their grievances and work issues.

a) Explain Toms research motivation type (4 Marks)
b) What research type is this? Explain (4 Marks)
c) Discuss one measurement scale which Tom must use in analyzing his variable (2 Marks)
d) Explain one method of data collection which will be appropriate for this research. Provide two reasons to justify your method (4 Marks)
e) What will be his appropriate data sampling technique? Justify your answer. (4 Marks)
f) Explain any four ethical considerations Tom should consider before staring his work (8 Marks)
g) In data analysis differentiate the range from standard deviation values (4 Marks)

Question 2: (20 Marks)

a) Explain two functions of a research hypothesis (4 Marks)
b) The below data was collected to investigate the spread of a certain virus infection among different animal species in a town. The data shows age of the animal who has been infected by the disease.

34	8	10	34
23	88	23	45
12	78	34	45
12	56	23	45
12	4	60	67
56	3	56	89
67	2	56	100
60	1	45	1
60	90	4	2
60	79	5	3
60	56	45	45

i. Calculate the Mean, Mode, Median, Standard deviation and range (10 Marks)
ii. Draw a histogram to represent the above data where the x axis should represent class boundaries and the y axis to represent the frequencies of distribution (6 Marks)

Question 3: (20 Marks)

You have just graduated and in a group of seven have decided to open company for developing software to clients. A client approaches you and would like you to develop an efficient mobile application. Answer the below question

i. Explain two sources of data collection for this project (4 Marks)
ii. Identify two possible concepts for this work and transform them into variables with a working definitions (12 Marks)
iii. Distinguish logical from empirical as characteristic of your research work (4 Marks)

Question 4: (20 Marks)

a) Distinguish between Conceptual vs. Empirical research types (4 Marks)
a) You are a research analyst and over the past five years you have been receiving awards as a best performer from your employer. You have been given the task of training new employees that have just been recruited by your company on best practices of conducting research. One of the learners would like to know the difference between nominal and interval scales of data rating, please explain. (4 Marks)
b) In your training sessions explain to the learners the following research methods (8Marks)

i. Cross-sectional Studies
ii. Ethnography
iii. Participative Enquiry
iv. Grounded Theory

c) State the difference between open and closed questionnaires (4 Marks)